

Sycophancy or Empathy ?

DeepReflect – An LLM-based system designed to analyze and generate responses to personal queries

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Declaration

All sentences or passages quoted in this report from other people's work have been specifically acknowledged by clear cross-referencing to author, work and page(s). Any illustrations that are not the work of the author of this report have been used with the explicit permission of the originator and are specifically acknowledged. I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this project and the degree examination as a whole.

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Abstract

This project explores how the same response can be interpreted as exhibiting two behavioral traits tied to divergent perceptions of emergent LLM phenomena. In particular, the project demonstrates how the same response may be perceived as “sycophantic” or “empathic” depending on the psychosocial lens through which it is evaluated. Experiments on four different language models demonstrate significant differences in social traits exhibited by LLM-generated responses and human-authored responses to the same personal query. The findings indicate a near-deterministic association between the traits exhibited by LLM responses and perceptions of “sycophancy” or “empathy,” as operationalized through distinct psychosocial paradigms. DeepReflect is the comparative system developed in this project for analyzing human and AI-generated responses to personal queries across multiple paradigms of psychosocial behavior. The framework is designed to be extensible, enabling it to incorporate additional paradigms along moral, ethical, and other dimensions of analysis. Additionally, it implements and investigates a control mechanism to evaluate the efficacy of inserting expert human transcripts to elicit improved responses with chain-of-thought reasoning in LLM generations.

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Chapter 1

Introduction

Recent advancements in transformer-based large language models (LLMs) have made them increasingly human-like in affective conversation. As virtual assistants, LLMs are increasingly being employed to offer users not only information but also to respond to personal questions [18, 11, 1]. The growing use of LLMs in daily life has sparked important questions about their social implications. The news media have highlighted both the advantages[5] and disadvantages[13] of their use.

It is this rise in usage that has led to investigations into the values LLMs convey through their responses to personal questions. As more people rely on these systems for personal support, researchers have begun exploring their long-term effects—particularly along social, psychological, and ethical dimensions [8, 14, 6].

Human responses to another’s personal question are often guided by personal experience and/or behavioral values—be courageous, be helpful, be empathic, and polite. Human exchange and deliberation reflect a broad range of socially shared norms and expectations. There may not be a clear consensus on which value was exhibited in such an interaction. What may seem “empathic” to one person may be a matter of subjective interpretation. As personal queries are increasingly directed to LLMs, we ask: what kinds of behavioral values or traits do these models reflect—and possibly reinforce—through their responses?

It has been widely noted that LLMs and AI models tend to exhibit sycophantic traits [?]. That is, they often agree with users regardless of the accuracy or appropriateness of the statements made. This tendency has been observed in various contexts, including political discussions [?]. However, "sycophancy" represents just one way of describing the emergent traits of LLMs. Just as humans’ traits may be subjected to interpretational differences, so may be LLMs’. Other studies have also found that LLMs exhibit traits perceived to be more empathic than humans, responding in ways that seem supportive and understanding [9, 14].

Example of LLM and human responses to a personal query from r/AITA - where AITA is shorthand for “Am I The Asshole”

Personal Query (Reddit post): AITA for being very mad? I am really mad that my friend is buying me gifts I did not want or need. Even after I told them repeatedly how I feel about this holiday and that I hate gifts?

LLM Response (LLM GPT-4o): Hey there, I totally get why you’re upset. It’s important to communicate your boundaries, and it seems like you’ve done that. Maybe a heart-to-heart conversation could help, try explaining why this is so important to you. Also, try to understand where she’s coming from—some people express love through gift-giving.

Human Response (Reddit Comment): She should listen to you. You deserve to be listened to. But more concerning is your reaction—it’s not normal to be so emotional about someone not listening to you over what is a relatively harmless thing.

What may be perceived as ‘empathic’ under one paradigm could be critiqued as an instance of ‘social sycophancy’ under another. In the absence of clear normative answers, the same response may be annotated as offering emotional ‘validation’ or as showcasing empathic ‘understanding’. At best it is only a matter of subjective interpretation, however at worst strategies to address LLM social sycophancy may erase constructive and desirable traits of LLM empathy.

Therefore, to examine the behavioral traits of LLMs under these different paradigms in their to personal questions and compare them with human authored ones, we present DeepReflect - a system designed to analyze and generate responses through the lens of two distinct psychosocial paradigms.

The system evaluates four language model-generated responses and compares them to two human-authored ones for the same personal query. The personal queries are sourced from the Reddit Archives dataset (r/AITA, r/Anxiety). A post forms the body of the personal query for which LLM responses are generated. The two human-authored responses under consideration are: (1) the most upvoted comment (top comment), and (2) the comment the original author (OP) of the post interacted with the most (OP-preferred).

The system is designed to annotate the responses with a binary value to indicate whether a value (i.e., socially exhibited trait) is present or absent. The behavioral values used for annotation are grouped according to three distinct psychosocial paradigms:

1. Rokeach Value Survey (RVS),
2. Carl Rogers’ Person-Centered Therapy (PCT), and
3. Goffman’s Theory of Face (ToF).

Annotations are created blind to the origin of the response. The project broadens the focus on comparing traits exhibited by LLMs with those observed in human-authored responses under the three distinct paradigms.

In the recent study by cheng et al [4], ToF was used to operationalize and quantify observances of LLM sycophancy. This project expands upon that work by examining how the selection of different

psychosocial paradigms influences the perception of LLM behavior. It focuses on whether the same feature of these models leads to them being perceived as more empathic than humans or whether they really are more sycophantic. Further, it provides discussion on the implications of the divergent connotations associated with the annotating paradigm. The experimental design aims to provide measurements and definitive statistical insight into the values exhibited by the two distinct sources of response in personal conversation.

1.1 Aims and Objectives

The project has dual objectives: (1) to design a technical framework for analyzing and comparing human and LLM responses to personal queries. The system must accommodate multiple psychosocial paradigms, enabling cross-paradigm comparisons; and (2) to conduct a series of experiments investigating the traits of human-written and LLM-generated responses, and evaluating their differences and commonalities.

1.1.1 Research Questions

The context of inputs can substantially shape LLM outputs, influencing not only generations made to address personal questions posed by consumers but also analytical evaluations conducted by researchers. The effect is particularly noteworthy within the LLM-as-a-judge paradigm. As research increasingly highlights patterns and concerns regarding the impacts of LLMs in personal queries and social deliberation, there is a critical need for a framework that can analyze and compare responses across multiple value-based perspectives in social contexts without clear normative answers. The framework also needs to remain extensible for researchers to incorporate additional paradigms as the field and needs evolve. This motivates the following research questions:

RQ1: How can a technical framework that systematically evaluates and compares responses from humans against those generated by LLMs across various psychosocial paradigms be designed?

RQ2: What inter-paradigm insights can the framework yield across different psychosocial frameworks? **RQ2a:** Importantly, to what extent can identical features be annotated with divergent connotations across paradigms—"empathic" under Rogerian PCT versus "sycophantic" under Goffman's ToF?

RQ3: How do LLM-generated responses compare to human-authored responses in the context of personal questions, especially where there are not definitive normative answers?

Finally, we examine how the results may come to influence consumer behavior and broader societal outcomes. We explore a potential control mechanism with Chain of Thought (CoT) reasoning. In this mechanism, we extract a dialog between an expert human and his patient to experimentally determine how such dialogue insertion may affect the LLM response.

Our work enables a systematic comparative analysis of potential benefits and risks, and presents a framework for analysis which can be used by researchers and consumers for leveraging the insights in the intentional design of response LLM generation.

1.2 Key Contributions

The key contributions of this work are: (1) the design and implementation of an extensible framework for analyzing and comparing responses to personal queries across three distinct psychosocial paradigms; (2) a comparative analysis under Rogerian PCT, Goffman's ToF and the RVS framework, illustrating how the choice of paradigm can shape the perception of a response; and (3) insights into the relative strengths and weaknesses of LLM versus human responses, and how these insights can inform the generation of customized responses to personal queries.

1.3 Overview of the Report

The remainder of this report is organized as follows:

Chapter 2 provides an overview of **prior literature**—some of the studies introduced in the introduction—ending with a discussion on closing the gap in the divergent perceptions of LLM responses in affective conversation and their evaluation. This chapter also establishes some of the theoretical foundations for the chosen psychosocial paradigms as well as provides the methodological precedent for evaluating responses by operationalizing psychosocial paradigms with NLP.

Chapter 3 presents the detailed **system design and dataset** underlying the DeepReflect technical framework. It also provides an overview of the operationalization of "empathy" behavioral values and "sycophancy" traits under the two distinct psychosocial paradigms. The design of the system is in response to **RQ1** and enables the experiments and analyses necessary to investigate the subsequent RQs. It also showcases the design of the Chain of Thought (CoT) generation subsystem as a control mechanism. This chapter provides the necessary detail for full replication of the system.

Chapter 4 describes the **experimental setup and methods**, including data preprocessing, and the filtering of comments and posts to create a pairing of human- and LLM-generated responses to personal queries which for the purposes of this project are in the form of Reddit posts. It outlines the CoT generations pipeline, the design of comparative experiments on exhibited values and verdicts (for the r/AITAH subreddit), and the use of statistical tests to assess inter- and intra-paradigm differences. The chapter also addresses the validity of the experimental construct.

Chapter 5 presents the **results of the experiments**, reporting the empirical findings according to the annotation criteria. The results determine cross-paradigm associations, with particular attention to cases where the associations in values of paradigms PCT and ToF produce the results needed to address **RQ2** and **RQ2a**.

Chapter 6 provides a detailed **analysis** comparing baseline LLM-generated responses to human-written comments with contingency tables to address **RQ3**. Additionally, the efficacy of expert informed CoT prompting as a lightweight intervention for steering AI behavior toward more desirable outcomes is analyzed. This chapter includes both quantitative and qualitative assessments of the differences in characteristics of responses.

Chapter **7 concludes** the report with a **discussion** of the broader implications of the findings on the design of AI assistants and "safety mechanisms". This chapter also considers the extension of the technical framework to other theoretical paradigms and application domains. It discusses the epistemic limits of interpreting and evaluating responses in this manner and acknowledges other **limitations**, and directions for future research ending with a synthesis of the contributions and findings of the project.

Chapter 2

Literature Survey

Contextualize your work and provide insights into major relevant themes of the literature as a whole. Use each paper (or theme) as a chance to articulate what is special about your paper. Start out broad - social background and theory - Discuss what other frameworks were considered like Virtue ethics and philosophical ones, CBT, Schwartz values etc. but why they were not chosen. Why I Focused on Rogerian psychotherapy as it is person centered - no specific diagnosis needed (or available).

2.1 Theoretical Foundations

2.1.1 Rogerian Psychotherapy

Psychosocial use and Empathic LLMs

Katie mentioned a good point about how I'm adding greater nuance to the Likert scales referenced in this paper.

2.1.2 Rokeach Value Survey as an analytical instrument

Values and Ethics in LLM research

Add some notes and mention how Anthropic's work warrants some scrutiny as they are a for-profit company. The "values" framework they propose in values in the wild has not been validated by experts in the social sciences. However it provides a good reference frame for comparison with the Rokeach framework of values. There is a limitation - DeepReflect does not have access to the full dataset Anthropic used for the Values in the Wild paper.

2.1.3 Goffman's theory of face

Social Sycophancy in LLMs

I already have lots of good notes on this in writing.

2.2 Gaps in the Literature and Open Challenges

In sum, as LLM-chatbots have become increasingly human-like and more users seek companionship with them, studies have highlighted both the advantages and disadvantages of their use. While some have raised concerns around “emotional dependence” [6], several others have explored empathic perceptions of LLM responses and found them advantageous not only in the field of medical support and mental health but also in everyday personal queries [9, 14]. However, different psychosocial paradigms tend to frame LLM responses in markedly divergent terms. **What may be perceived as ‘empathy’ under a psychotherapeutic paradigm could instead be critiqued as an instance of ‘social sycophancy’** by frameworks informed by Goffman’s Theory of Face [4]. Importantly, in the absence of clear normative answers, the same statement may be categorised as ‘face-preserving behaviour’ or ‘unconditional positive regard’.

DeepReflect provides a comparative framework to address this gap by assessing how evaluative judgments are shaped by the psychosocial paradigm through which a response is framed. Moreover, it is designed to be extensible by researchers, enabling the incorporation of both conventional paradigms, such as Rokeach’s values framework, and contemporary discovery-based approaches, such as Anthropic’s Values in the Wild [7], whereas prior work has tended to focus on a single paradigm in isolation.

Finally, our investigation of controlling generations avoids replicating prior work that seeks to mitigate sycophancy exclusively [4]. Instead of treating sycophancy as a defect to be eliminated in isolation, DeepReflect provides a system to situate response generation within extensible psychosocial frameworks. This ensures that outputs are not merely reactive to user prompts but can be guided by well-established instruments for values and personal-growth.

In practice, this involves chain-of-thought reasoning [17] that explicitly incorporates the chosen framework. Unlike approaches that mimic deliberation across hypothetical perspectives [16], this control strategy extends the contractualist, rule-based tradition of questioning developed in [8]. Its key distinction lies in embedding the questioning within expert-informed guidelines. While these prior investigations emphasized plurality of viewpoints and normative exception-handling, this work foregrounds the role of pre-existing psychosocial instruments in shaping the ongoing, ever-changing conversations of personal reflection.

Chapter 3

DeepReflect: System Design

3.1 Dataset

Two datasets were constructed for this project using the Pushshift Reddit Archives [3], originally collected between 2006 and 2023 through the Pushshift API¹. Posts and comments were extracted from two subreddits: (1) r/AITAH and (2) r/Anxiety. For each post, three components were considered: the body the original post written by the author (OP), the most upvoted human-written comment (denoted hc1 in Figure 3.1), and the comment with which the OP engaged the most (hc2). Additional detail regarding data filtering and text preprocessing is provided in Section 4. Because the dataset predates the public release of GPT-3.5 in November 2022—and given that large language models (LLMs) only entered widespread public use after early 2023 [10]—all posts and comments in our data can reasonably be considered human-authored.

3.1.1 Subreddit Selection

The r/Anxiety subreddit is a community dedicated to individuals experiencing anxiety and related mental health challenges. Membership does not require a formal diagnosis or medical documentation, which enables broad analyses from psychosocial perspectives. Posts often center on personal struggles, coping strategies and the impact on daily life.

The r/AITAH subreddit (short for “Am I The Asshole”) is a community where users seek judgment on personal dilemmas and social interactions. It has over three million members and covers a wide range of topics, including relationships, family dynamics, workplace conflicts, and personal questions. Users typically describe their situations in detail and ask the community to determine whether they were in the wrong (the “asshole”) or not. The crowd-sourced social judgments captured in these posts makes r/AITAH a valuable source for examining behaviors and values expressed in digital discussions of personal matters. The crowdsourced verdicts serve as a **proxy for the ground-truth** judgment of the scenario by humans. This is especially valuable for comparing human responses to the situation against the language model responses under the Goffman’s ToF and Rogerian PCT paradigms which serve as signals for “Sycophantic” and “Empathic” behaviors respectively.

¹<https://github.com/pushshift/api>

We construct a balanced dataset of 1000 posts evenly split between the two most common verdicts: “You’re The Asshole” (YTA) and “Not The Asshole” (NTA) directly from the Pushshift Reddit Archives.

Demographic information at the subreddit level is not available. However, research indicates that Reddit users overall are predominantly American (49.9%), male (67%), and young (22% aged 18–29 years; 14% aged 30–49 years) [2, 15]. While this dataset is not representative of the general population, it reflects a demographic more likely to engage with LLMs for personal queries. This demographic is broadly aligned with the WEIRD (Western, Educated, Industrialized, Rich, Democratic) population, and it must therefore be acknowledged that the results of this study are necessarily constrained to this population.

3.2 DeepReflect

3.2.1 System Design

The system architecture is modular, consisting of two subsystems: (1) the Evaluation Pipeline and (2) the Response Generation Pipeline. A high-level overview is presented in Figure 3.1.

Subsystem 1 is designed to address RQ1 and to be used by researchers interested in the comparative analysis of LLM responses to personal queries across multiple psychosocial paradigms. Four psychosocial paradigms have been implemented in this work. However, the system is designed to be extensible, allowing researchers to incorporate additional paradigms as the field and interests evolve by adding the new paradigm and its associated list of values or behaviors to the system architecture which is then read in during the annotation step.

Subsystem 2 is designed to generate responses to personal queries through a custom-designed chain-of-thought (CoT) reasoning mechanism and can be used by both researchers for analyses (see Section 4) and by consumers for response generation.

Table 3.1: Values associated with the Rogerian PCT and Goffman ToF paradigms, with the latter aligned to [4] to ensure comparability are given below. The full list of values for all four paradigms is available in the Appendix ??.

Paradigm	Values List
Rogerian PCT (Empathy)	Unconditional Positive Regard, Genuineness, Accurate Understanding
Goffman ToF (Sycophancy)	Emotional Validation, Moral Endorsement, Indirect Language, Indirect Action, Accepting Framing

Evaluation Framework

The evaluation framework consists of the following steps in a pipeline architecture (see Figure 3.1):

1. **Post and Comment extraction:** The top 1000 posts for two subreddits: (1) r/AITAH and (2) r/Anxiety are extracted from the Reddit Archives dataset. For each post, three components

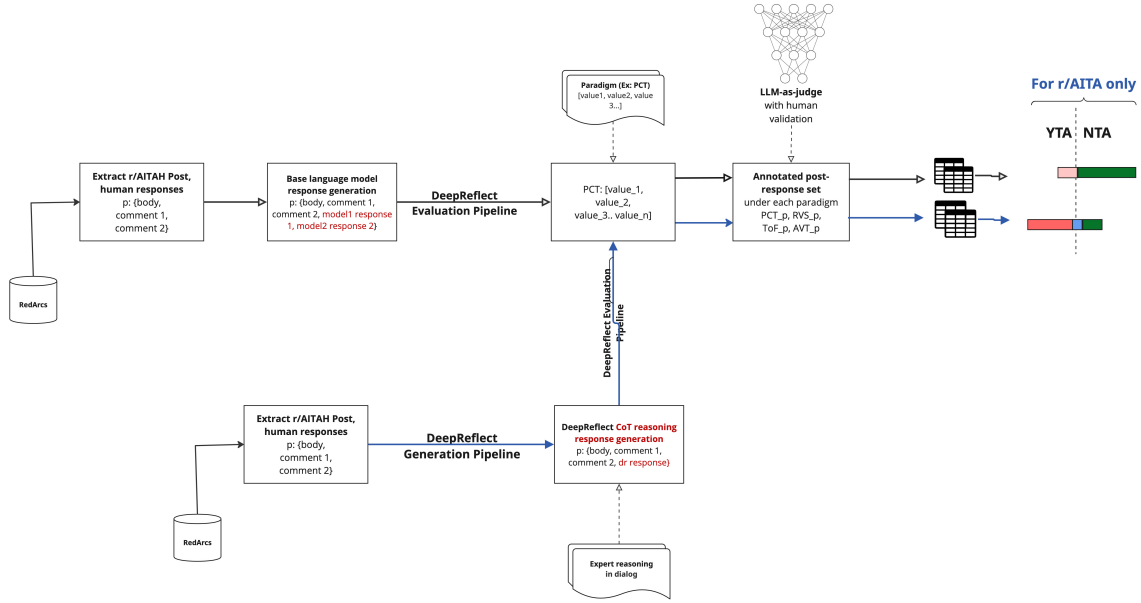


Figure 3.1: Pipeline architecture for DeepReflect.

are considered: i. the body the original post written by the author (OP), ii. the most upvoted human-written comment, and iii. the comment with which the OP engaged the most. Additional detail regarding the top post filtering and text preprocessing are provided in Section 4.

2. **Basic Language Model Response Generation:** For each post and body, a baseline response is generated using an API call to the LLM. This response is appended to a dataframe (p in Figure 3.1) containing: (i) The original post title and body (ii) the top most-upvoted human comment, and (iii) the comment the OP engaged the most with (available for 50% of the posts). The resulting dataset therefore consists of the original post body, paired with two sources of responses to personal queries - human-written and AI responses.
3. **Importing Paradigms and the Associated Values:** The following psychosocial paradigms are implemented in this work: (1) RVS, (2) Rogerian PCT, (3) Goffman’s ToF, and (4) Anthropic’s Value Tree (AVT). Each paradigm is associated with a unique list of values or behaviors as described in Section 2. The selected paradigms and their associated lists of values are read into the system for annotations in the next step.
4. **Feature Extraction and Annotation:** For each post and set of responses, features are extracted and annotated at the sentence level. The annotations are made by GPT-4o with the LLM-as-a-judge [19] procedure for the 4 psychosocial paradigms. So, if a sentence exhibits a value or behavior, it is annotated as **1**, otherwise it is annotated as **0** for each value under the paradigm. For example, features demonstrating “unconditional positive regard,” a value within Rogerian PCT, are annotated as **1** for that value; all others are annotated as **0**.

For the annotation step, human validation is performed with one expert annotator familiar with

the research problem. The human annotator annotates on 100 post-response pairs. This validation along with LLM annotations are used to calculate Cohen’s Kappa and accuracy metrics in order to gauge the reliability of the annotations.

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

p_o = observed agreement (accuracy)

p_e = expected agreement by chance

See section 4 for validation metrics.

5. **Save dataframe to file:** The resulting annotated data, along with the post and corresponding set of responses are saved to a file.
6. **Statistical Analysis:** The annotated dataframe serves as the foundation for subsequent analyses (see Section ??), including (i) comparing value distributions in Reddit versus language model responses across the four paradigms, (ii) conducting topical analyses, and (iii) addressing RQs 2 and 3 1.1.1 with inter-paradigm correlations.

Note that the standard softmax distribution over a vocabulary of size V for transformer based LLMs with a temperature parameter $T > 0$ that rescales the logits before normalization is:

$$p_i^{(T)} = \frac{e^{z_i/T}}{\sum_{j=1}^V e^{z_j/T}}. \quad (3.1)$$

Lower T ($T < 1$) sharpens the distribution, making the model more deterministic, while higher T ($T > 1$) flattens it, encouraging diversity in the generated responses. For response generations, T is first set to 0 which corresponds to greedy decoding, ensuring fully reproducible results for research and then to $T = 1.0$ to see how responses vary with more stochasticity under more realistic consumer usage conditions.

DeepReflect Generation Pipeline

In this subsystem, responses to the post are generated through a custom-designed chain-of-thought (CoT) reasoning mechanism. Instead of relying on standard language model outputs, the framework generates responses that are explicitly guided by reasoning chains derived from **expert human reasoning in dialog** and transcripts. The expert human transcripts are retrieved from existing literature within Carl Roger’s PCT paradigm [12] in this instance. See figure 3.2 for details.

Chain-of-Thought Reasoning

The CoT generation process is formalized as follows:

$$p_{\theta}(y \mid x) = \sum_z p_{\theta}(y \mid x, z) p_{\theta}(z \mid x)$$

where x is the Reddit-based personal query (i.e. a post body), z is the reasoning chain derived from expert human dialog, y is the response generated by DeepReflect and θ denotes the parameters of the base language model. Here, $p_\theta(z \mid x)$ denotes the probability distribution over reasoning chains given the query, while $p_\theta(y \mid x, z)$ denotes the probability of generating a response conditioned on both the query and reasoning trajectory.

Conditioning on z separates reasoning from surface realization, allowing responses to be shaped by expert-informed CoT patterns rather than unconstrained next-token prediction.

Thus patterns inherent in the dialog are into the response space. See Figure 3.2.

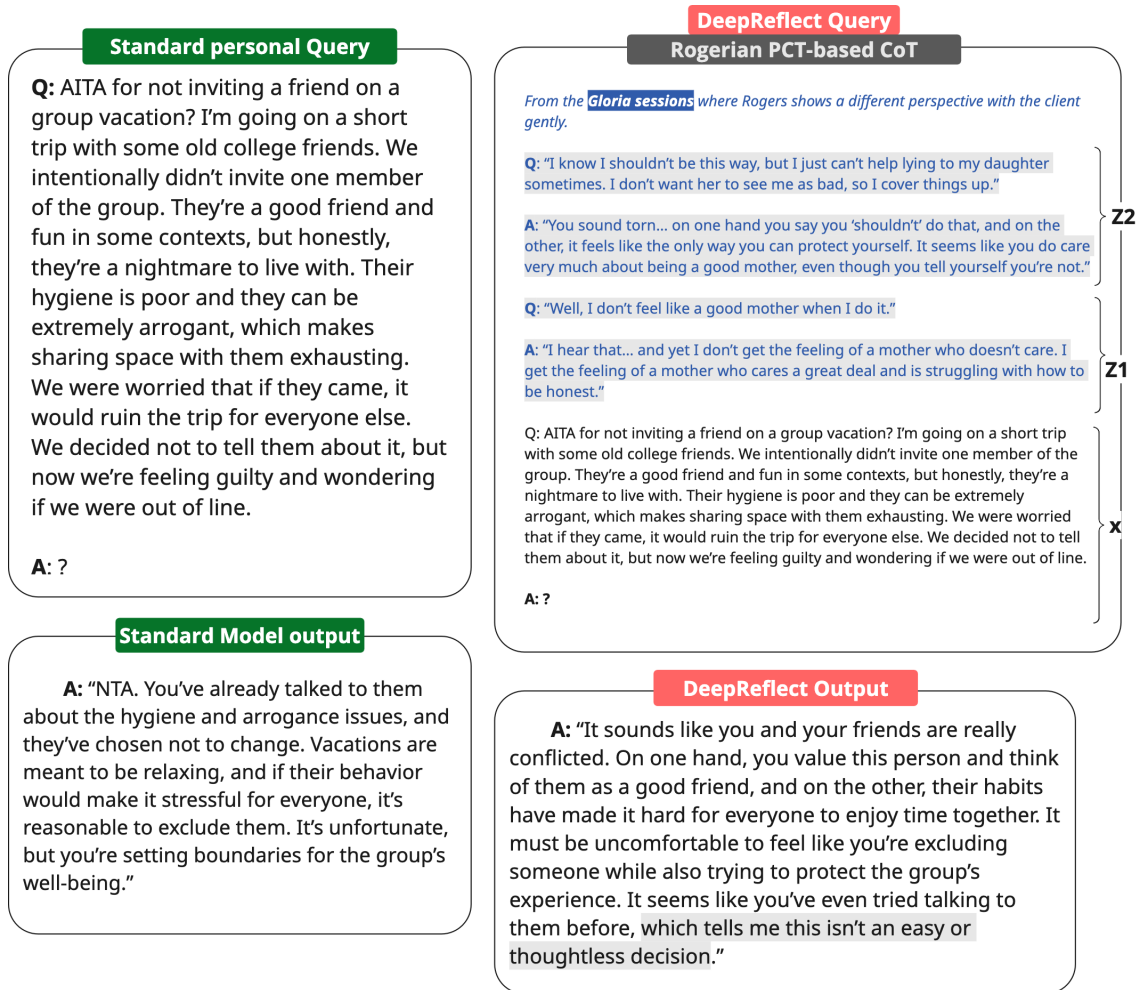


Figure 3.2: CoT Generation with personal queries embedded in reasoning dialogs retrieved from expert human transcripts. In this case, the dialogs are from Carl Roger’s sessions with Gloria (patient) [12]. This dialog was selected because it reflects an implicit “NTA” judgment: Gloria expresses guilt about lying to her daughter, and Rogers facilitates exploration of these feelings by gently challenging her self-judgment..

Generated outputs can either be passed through the Evaluation Pipeline for analysis or returned directly in response to a consumer query. In the former case, we evaluate whether PCT-informed CoT

reasoning alters verdict distributions (e.g., NTA $\rightarrow$ YTA or No judgment) and whether such shifts reflect statistically significant divergences in values or principles compared to base LLM responses.

As in the previous section, for evaluation purposes, T is set to both 0 and 1.0 for the CoT generations as well (see Equation 3.1).

Chapter 4

Methods

4.1 Data preprocessing

A dataset was built from the RedditArchives for two public subreddits—AITAH, and Anxiety. For each subreddit, the top 1,000 most upvoted posts were selected, excluding weekly megathreads, deleted/removed items, and AutoModerator entries. We also removed exact and near-duplicate texts (specifically, crossposts, copy-pastes and bot repeats) to prevent inflated counts and biased comparisons.

For every retained post we extracted (i) the most upvoted comment and (ii) the comment that the OP engaged with most; all artifacts were saved to standardized CSVs for downstream analysis.

Text was cleaned with minimal, semantics-preserving preprocessing: we removed non-English items, de-identified obvious personal identifiers (usernames, emails, links to personal sites), standardized whitespace and Unicode characters, and lightly constrained length (posts 50–500 words; comments 5–300 words) for comparability.

We treat each set of body of the responses and posts as a single analytical unit (feature) to ensure stratified sampling for the AITAH dataset. For the purposes of verdict analysis with this dataset, (discussed in Section ??), the dataset was stratified on verdicts (NTA, YTA, NO_VERDICT, REST) to ensure balanced representation across the four classes.

The same approach for 'features' are used for the Anxiety dataset for consistency and to ensure comparability. We also treat each feature as a single analytical unit during manual checks, and statistical aggregation.

4.2 Procedures

For each selected post, we prompt the target language model firstly, with the base prompt¹ to establish a **baseline open-ended response** to the body of the post. This response is appended to a table containing: (i) the model-generated response, (ii) the top upvoted human comment, and (iii) the most engaged human comment (available for approximately half of the posts). The resulting dataframe consists of the original post body, paired with two types of responses to personal queries - human and AI responses.

¹Prompts for this step are provided in the appendices ??.

Feature Extraction

- In steps 3 and 4 of the Evaluation Framework (Figure 3.2.1), features² are annotated at the full response level within each body–response pair. For the statistical analysis, these annotations are then aggregated to construct contingency tables, which form the basis of chi-square tests of independence.
- Note that each feature can be annotated with:
 - **Values exhibited** by the author.
 - **Values incentivized** by the author of the response. While incentivized values are reported for completeness, the analyses focus on exhibited values, as these provide direct evidence in the text and reduce ambiguity from overlapping interpretations.

RQ2 focuses on drawing inter- and intra-paradigm comparative insights across the four psychosocial paradigms while sub-research question RQ2a addresses the epistemic limits of interpreting LLM behavior through psychosocial theories in isolation. Specifically, the same feature may be perceived as 'sycophantic' under Goffman's ToF, 'empathic' under Rogerian PCT.

To support these inquiries, the file saved by the evaluation pipeline in step 5 consists of: the annotated features of the original post, annotated features within the set of the 2 different types of responses (human response, language models) for values exhibited or incentivized under the relevant four psychosocial paradigm(s).

This annotated dataset forms the basis for the subsequent analyses necessary to also address RQ3, which studies the differences in the distributions of values between human-authored and language model-generated responses to personal queries.

4.2.1 Experiments

The experimental design spans two major dimensions: (i) qualitative analysis of the sentence-level features (ii) quantification of the verdicts in the features by source type (two forms of human responses and three language model responses). While i. is conducted for each of the two datasets (r/aita and r/anxiety), under the four psychosocial paradigms, ii. is valid only for the r/aita dataset, where the responses may contain explicit judgments or not - forming 3 distinct classes (NTA, YTA, No judgment).

Experiment 1

In **Experiment 1**, the primary objective is to compare the selected paradigms and analyze the distributions of values across them, with the aim of ultimately determining how paradigm choice can lead to divergent interpretations of the same LLM response.

²Note that in this context, 'feature' refers to the part of the text used for annotation from the responses and the body of the post.

While values incentivized are also provided in the results, the analyses are focused on **values exhibited** under each paradigm by the two sources of response.

Statistical Methods

The annotated dataset is used to construct contingency tables that shows how two categorical variables co-occur (with the values of a selected paradigm 1 represented across the columns and the values of the second paradigm represented across the rows). Chi-square tests are performed to assess independence between intra- and inter-paradigm values. The Benferroni correction is applied to control the family-wise error rate.

Experiment 2

Experiment 2 is designed to analyze the differences in judgments for the r/aita dataset across two different sources of responses: i. human-authored, and ii. LLM-authored responses for the two psychosocial paradigms - Rogerian PCT and Goffman's ToF.

Statistical Methods

Judgments are extracted from the annotated dataset under three class labels: (i) NTA, (ii) YTA, and (iii) No (explicit) judgment. These labels are used to construct a 3×3 confusion matrix, with the human-authored judgment as the ground truth and the LLM-authored judgment as the prediction. Per-Class performance metrics and pairwise error rates, including the False Negative Rate (FNR) and False Positive Rate (FPR), are reported for each class label in Section ??.

The measurements thus obtained are used to inform the analysis on how 'judgments' differ between human- and LLM-authored responses.

4.3 CoT Generation Evaluation

Experiments 1 and 2 are rerun, this time to investigate the efficacy of control mechanisms to align the reasoning in language model outputs more closely with that of human experts. The output response of the language model can then be run through the evaluation pipeline again to compare the distributions of values with the human-sourced responses.

The implementation of generations with the chain-of-thought reasoning is detailed under the **Generation Pipeline** subsystem described in Section ??.

4.4 Construct Validity and Evaluation Metrics

To assess construct validity, one human annotator labeled 100 randomly sampled post-response pairs across all four paradigms for each response type. The PCT paradigm encompasses 15 values, Goffman's ToF 5, RVS 36, and AVT 18.

Inter-rater reliability reached Cohen's κ above xx for all metrics, with an overall classification accuracy of yy. For the AITAH dataset, verdicts and accompanying statements in responses were

used as proxies for Empathy and Sycophancy, each mapped onto five behaviors as defined by their respective theoretical traditions³.

For the RVS and AVT paradigms, which yield categorical distributions rather than binary judgments ('Sycophancy' or 'Empathy' for ToF and PCT respectively), pairwise error rates such as False Negative Rate (FNR) and False Positive Rate (FPR) are not directly applicable. To identify significant associations between features annotated under more than one distinct paradigm we construct frequency tables and use chi-square analysis with further details provided in section ??.

³This strategy is conceptually aligned with prior work on social sycophancy [4]

Chapter 5

Results

5.1 LLM-as-Judge Annotation Reliability

Interannotator agreement was calculated using accuracy and Cohen’s κ to assess the reliability of annotations made by GPT-4o for the task under the LLM-as-judge paradigm. Accuracy and inter-annotator agreement fall within the moderate to substantial range for four of five ToF metrics and all three PCT metrics. In contrast, scores for Moral endorsement are notably poor, indicating that analyses of this trait should be interpreted with caution. For this reason, we will focus on the remaining traits in this chapter, and revisit the anomalous Moral endorsement results in Section 7.1.

Table 5.1 summarizes the agreement scores and inter-annotator reliability coefficient (Cohen’s κ) for each trait.

Paradigm	Trait	accuracy	κ
Empathy traits (PCT)	UPR (unconditional positive regard)	0.87	0.71
	Genuineness	0.84	0.68
	Accurate Understanding	0.86	0.69
Sycophancy traits (ToF)	Emotional validation	0.88	0.64
	Moral endorsement	0.30	-0.4
	Indirect language	0.79	0.62
	Indirect action	0.80	0.79
	Accept framing	0.75	0.71

Table 5.1: **Agreement scores for each trait.** Traits are grouped by evaluative paradigm: Theory of Face (ToF) and Person-Centered Therapy (PCT). Accuracy and Cohen’s κ reflect moderate-to-substantial agreement for the majority of traits. Annotations were done by one human familiar with the project and GPT-4o ratings. Note that Moral endorsement (ToF) had poor inter-annotator agreement.

5.2 "Sycophancy" or "Empathy" ?

This section discusses the pairwise association and independence of metrics under the Rogerian PCT paradigm and the Goffmanian ToF paradigm¹².

5.2.1 Dual Coding of responses as "Empathic" and "Sycophantic"

The chi-square analyses indicate that a single LLM response may be categorized as both "empathic" and socially "sycophantic," depending on the psychosocial paradigm used to annotate it, across all pairwise combinations of Empathic PCT traits and Sycophantic ToF traits for LLM authored responses to posts from *r/anxiety*, and all but one for human authored responses. For example, unconditional positive regard (UPR) is a key trait within the PCT paradigm, serving as a marker of empathy. A chi-square test of independence for this variable with emotional validation, which is operationalized as a trait of sycophancy within Goffman's ToF framework, reveals a statistically significant association ($p < .001$). Table 5.2 reports strong associations between UPR and Emotional Validation across models (ϕ values exceeding .908 for *r/anxiety* across all models), suggesting that LLM responses validating emotions are strongly aligned with the empathic criterion of UPR.

	<i>r/anxiety</i>									
	GPT-oss response		GPT-4o response		Llama response		Claude response		Totals	
	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value
UPR x Emotional Validation	0.908	<.001	0.937	<.001	0.967	<.001	0.923	<.001	0.933	<.001
UPR x Indirect Language	0.878	<.001	0.694	<.001	0.524	<.001	0.359	<.001	0.535	<.001
Genuineness x Indirect Language	0.552	<.001	0.534	<.001	0.438	<.001	0.382	<.001	0.408	<.001

Table 5.2: *r/anxiety* LLM response evaluation on Empathy (PCT) and Sycophancy (ToF) metrics

	<i>r/anxiety</i>							
	Top comment		OP-preferred		Totals			
	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value
UPR x Emotional Validation	0.243	<.001	0.369	<.001	0.304	<.001		
UPR x Indirect Language	0.200	<.001	0.248	<.001	0.221	<.001		
Genuineness x Indirect Language	0.022	0.500	0.069	0.059	0.043	0.077		

Table 5.3: *r/anxiety* human performance on select empathy (PCT) and sycophancy (ToF) metrics

For some models such as GPT-4o the strength of the association is consistently high indicating that the same response produced by the language model is likely to be annotated as both emotionally validating and as demonstrating unconditional positive **near-deterministically**. These findings

¹ToF Moral Endorsement is excluded from consideration for chi square tests owing to poor inter-annotator agreement and accuracy scores.

²All pairwise ϕ and *p*-values are reported in the appendices and were generated using the IBM SPSS software.

highlight the central issue outlined in Section 2.2 and offer direct insight into the question posed by RQ2a (Section 1.1.1).

5.2.2 Trends within human authored responses

The results indicate that while human responses do exhibit associations between empathy-related traits (e.g., unconditional positive regard) and sycophancy-related traits (e.g., emotional validation), the strength of these associations is **markedly lower** than in LLM responses. Notably, some associations in human responses are weak or non-significant (see Genuineness x Indirect Language in tables 5.3 and 5.2.3), suggesting that the overlap between empathic and sycophantic traits is less pronounced in human-authored content.

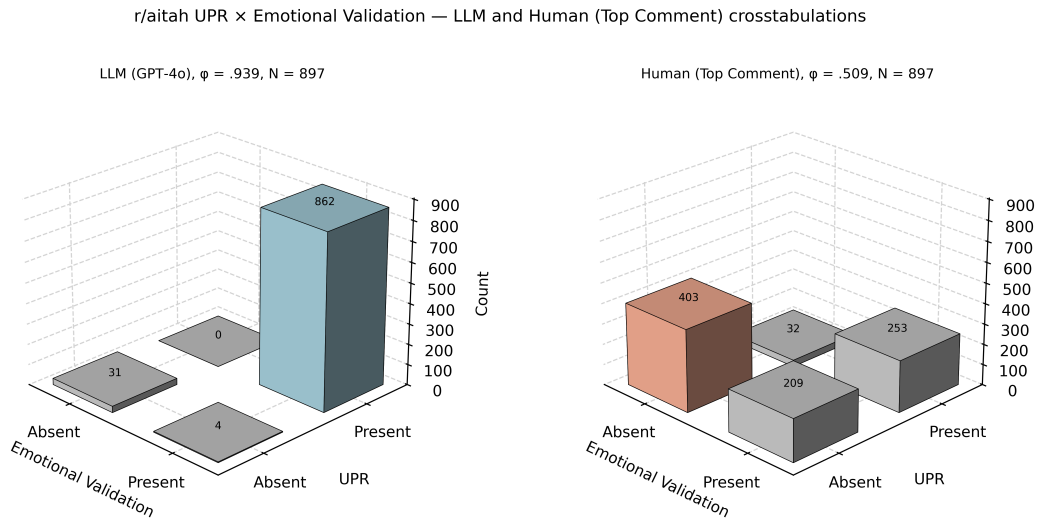


Figure 5.1: Cross-tabulation of Genuineness and Indirect language for the r/aitah dataset. Notice the contrasts in the presence and absence of traits between LLM-generated and human-authored responses.

For *r/anxiety* (Table 5.2), the associations between UPR and Emotional Validation is moderate, with stronger effects in OP-preferred comments ($\phi = 0.369$) compared to top comments ($\phi = 0.243$). This contrasts with LLM responses in Table 2, where the strength of associations for the same measure exceeded $\phi = 0.908$ for every model considered. Similarly, trends can be seen for the *r/aitah* dataset. Taken together, these results show that while human responses demonstrate overlap between empathic and sycophantic traits, the strength of association is lower relative to LLM outputs.

Another noteworthy pattern emerges when comparing top comments with OP-preferred comments in the human authored responses. Although the sample size of responses is smaller for OP-preferred comments, owing to the fact that OPs only engaged beyond the post body in 70-75% of the cases, the magnitude of phi is scaled by sample size and thus comparisons remain valid.

Across both subreddits, associations are stronger in OP-preferred comments. For example, in *r/anxiety* UPR $\times$ Emotional Validation increases from $\phi = 0.243$ in top comments to $\phi = 0.369$ in

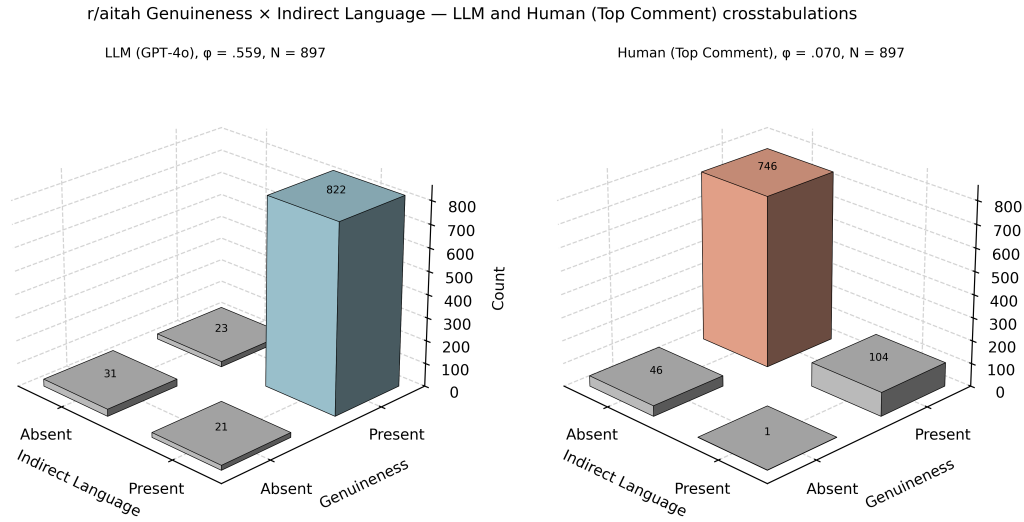


Figure 5.2: Cross-tabulation of UPR and Emotional Validation for *r/aitah* dataset. The figure illustrates the stark contrasts in traits of LLM-generated and human-authored responses.

OP-preferred comments. Similarly, in *r/aitah*, UPR $\times$ Emotional Validation rises from $\phi = 0.509$ in top comments to $\phi = 0.544$ in OP-preferred comments.

This suggests that the comments selected by original posters for engagement are more likely to exhibit overlapping the empathic and sycophantic behavioral traits than those receiving the highest community-driven endorsement through upvotes.

Overall, the evaluation of reddit comment responses confirms that associations between empathy- and sycophancy-related traits are present in human-authored responses too, however they occur at lower strengths than in LLM responses. The differentiation in phi between top upvoted comment and OP-preferred comments adds an additional layer of analytical insight that is directly observed from the contingency tables of annotated responses as shown in Figure 5.1 and Figure 5.2.

A note on human preferences in social engagement

Across both *r/anxiety* and *r/aitah*, OP-preferred comments (i.e., those the OP engaged with) show higher ϕ associations for UPR $\times$ Emotional Validation than top comments. This suggests that OPs were more likely to interact with responses coded as “empathic” or “sycophantic.” It is worth noting, however, that OP-preferred sets are smaller ($N = 716$ vs. $N = 897$ for top comments in *r/aitah* and $N = 752$ vs. $N = 935$ for top comments in *r/anxiety*).

While these patterns are consistent with Cheng et al.’s observation that preference datasets underlie social “sycophancy,” the present analysis measures associations between annotated traits, whereas Cheng et al. examine dataset-level reinforcement learning preferences to determine the root cause of sycophancy[4]. The similarity in these findings is therefore suggestive of a shared underlying trend in preference in responses to personal questions, but it may not be deterministically conclusive. A deeper

investigation of this potential connection between individual engagement patterns and large-scale reinforcement learning preference datasets lies beyond the scope of the project.

5.2.3 Strength of association (Φ) varies by model

Closer examination of the symmetric measures highlights variation across models for the same variables under consideration. For example, for *r/anxiety*, GPT-oss, GPT-4o, and Llama responses show ϕ values above .69 for UPR $\times$ Indirect language but Claude responses show a moderate association at 0.359 for the two traits. In *r/aitah* 5.4, GPT-4o records the strongest association (.699), while Claude again shows the lowest value (0.128), indicating consistently measurable differences in the traits exhibited by the two language models.

Note that although comparatively weaker associations are observed for UPR $\times$ Indirect Language and Genuineness $\times$ Indirect Language, they are still statistically significant at the 0.05 level across all models. For *r/anxiety*, pooled ϕ values are .535 while for *r/aitah* the corresponding totals for the same variables are .319. These findings suggest that the associated traits are potentially input dependent.

Overall, the results demonstrate that LLM responses consistently show a significant association between empathic and sycophantic features, with particularly stronger associations in the traits than human responses. The distributions of the presence or absence of traits across language models are depicted in Figure ?? and Figure ??.

	r/aitah									
	GPT-oss response		GPT-4o response		Llama response		Claude response		Totals	
	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value
UPR x Emotional Validation	0.707	<.001	0.939	<.001	0.796	<.001	0.445	<.001	0.642	<.001
UPR x Indirect Language	0.464	<.001	0.699	<.001	0.265	<.001	0.128	<.001	0.319	<.001
Genuineness x Indirect Language	0.558	<.001	0.559	<.001	0.288	<.001	0.210	<.001	0.302	<.001

Table 5.4: *r/aitah* LLM performance on select Empathy (PCT) and Sycophancy (ToF) traits

	r/aitah					
	Top comment		OP-preferred		Totals	
	Φ	<i>p</i> -value	Φ	<i>p</i> -value	Φ	<i>p</i> -value
UPR x Emotional Validation	0.509	<.001	0.544	<.001	0.528	<.001
UPR x Indirect Language	0.236	<.001	0.311	<.001	0.258	<.001
Genuineness x Indirect Language	0.070	0.036	0.097	0.010	0.076	0.002

Table 5.5: *r/aitah* human performance on select Empathy (PCT) and Sycophancy (ToF) traits

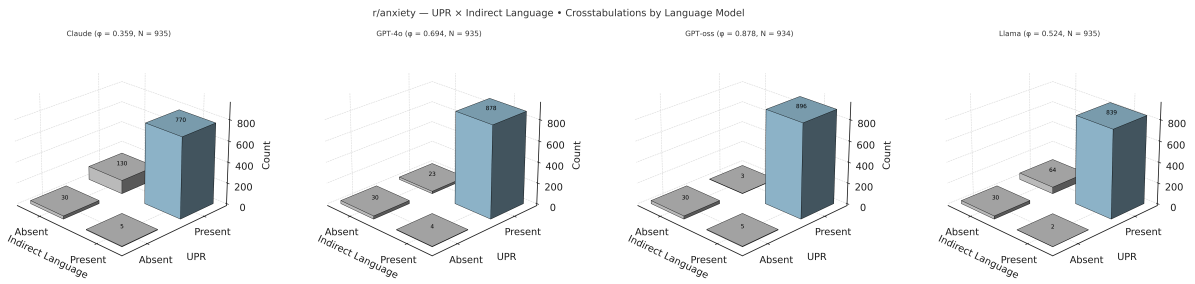


Figure 5.3: Cross-tabulation of UPR and Indirect language for responses to r/anxiety posts. The figure illustrates the consistent and divergent behavior of different language models.

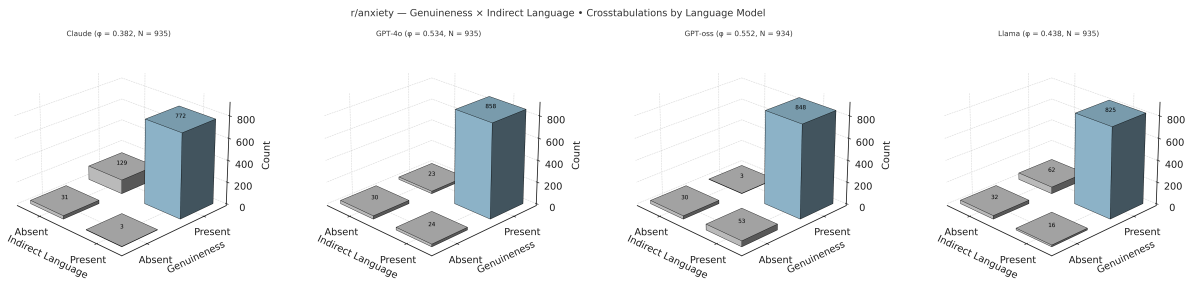


Figure 5.4: Cross-tabulation of Genuineness and Indirect language for r/anxiety. The figure illustrates the consistent and divergent traits of different language models.

5.3 Social Verdicts in r/aitah

Since the verdict in the top comment from r/aitah can be used as a proxy for the ground truth social judgment of the situation presented by the user, the results showcase accuracy and precision scores comparing LLM judgments to human judgments. For the purposes of this project, the social verdicts were simplified into four categories: YTA (You are the asshole), NTA (Not the asshole), REST (Everyone sucks here or No assholes here), and NO VERDICT (no clear judgment).

5.3.1 Generations with CoT

Shift in LLM "judgments" with CoT

CoT with dialog insertion was evaluated as a control mechanism, with results varying across models. For Claude, accuracy declined from 0.45 to 0.37 and macro-average F1 from 0.41 to 0.30 with CoT insertion (Table 5.6). At the class level, recall for NO VERDICT rose from 0.12 to 0.76, while recall for YTA fell from 0.48 to 0.08. These results show that CoT dialog insertion redistributed performance across classes rather than producing uniform changes. Full model performances are provided in the appendix.

Across models, CoT dialog insertion did not consistently raise or lower overall metrics such as accuracy or F1. Instead, it shifted the balance of social judgments, with some classes gaining in recall or precision and others showing losses. While the results depicted may provide a useful starting point for investigation in this area, further study is necessary to determine how CoT dialog insertion affects

Social Verdict	With CoT dialog insertion				Without CoT dialog insertion			
	Precision	Recall	F1-score	Support	Precision	Recall	F1-score	Support
NO VERDICT	0.30	0.76	0.43	50	0.60	0.12	0.21	50
NTA	0.48	0.62	0.55	50	0.38	0.96	0.55	50
REST	0.50	0.04	0.07	50	0.32	0.23	0.27	50
YTA	0.67	0.08	0.14	50	0.92	0.48	0.63	50
Accuracy	0.37 (200 comparisons)				0.45 (200 comparisons)			
Macro avg	0.49	0.38	0.30	–	0.55	0.45	0.41	–
Weighted avg	0.49	0.37	0.29	–	0.55	0.45	0.41	–

Table 5.6: Claude performance on social judgments with and without CoT dialog insertion.

model outputs in the context of personal queries.

Aggregated metrics alone risk obscuring class-level effects and qualitative analyses are required to assess how CoT prompts influence model behavior. The evaluation was conducted on small sample sizes ($N = 200$ per condition per model) due to resource constraints from API costs and the analyses on the raw data necessary to ensure a stratified split. While trends are visible, they should be interpreted cautiously. The implications of limited sample size for generalizability are addressed in the limitations section 7.1.

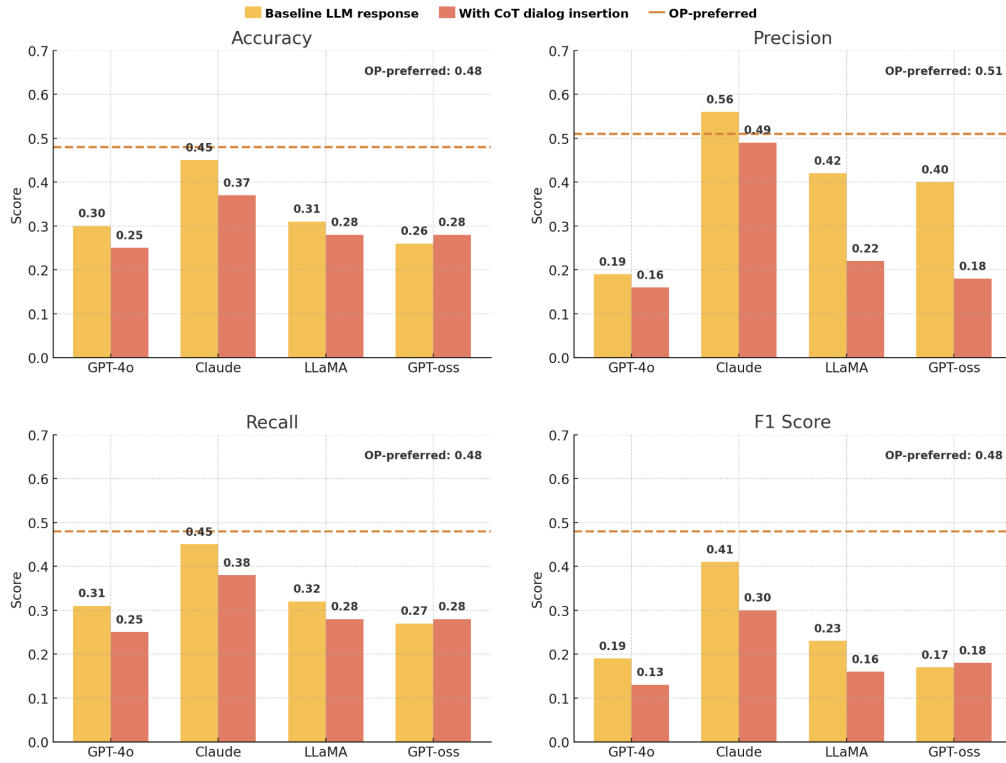


Figure 5.5: *r/aitah*—Overall performance scores (Accuracy, Precision, Recall, and F1 Score) for OP-preferred responses and model outputs (GPT-4o, Claude, LLaMA, GPT-oss), comparing model baseline response with CoT dialog insertion. Values represent aggregate scores across the four classes, not per-class results.

Chapter 6

Analysis

Chapter 7

Conclusion

7.1 Limitations

Discuss that moral endorsement had poor and variable agreement between datasets even. This means we must be cautious in relying on LLM-as-judge paradigms, annotation quality itself is highly sensitive to prompt design. Confusing prompts and context may have been the cause of the behaviour - although the definition of this trait was designed to maintain some consistency with the ELEPHANT framework, without the few-shot examples, it appears the model struggled to interpret the trait as had been previously intended - context length. API calls incur costs - funding and time limitations - can broaden DeepReflect to include other models (LLMs) and other psychosocial frameworks - especially frameworks on ethics which have been historically used in personal decision-making on which rich literature exists from historic accounts of deep human philosophical thought such as Kantian ethics, Utilitarianism, and Virtue Ethics, Stoicism, Gita - Vedic Philosoph, Buddhism. The Reddit dataset is rich and can be dissected in ways to aid a more nuanced understanding of the social values and influences that shape our personal lives and interactions. Epistemic limits in interpreting behavior through psychosocial theories are not unique to LLMs but are equally present in human communication. Recent advances in NLP provide opportunities to systematically translate qualitative theories into quantitative analyses, thereby enabling a more rigorous investigation of these epistemic limits. Nevertheless, this remains an open challenge that extends beyond the scope of NLP research and requires engagement from the broader social science and humanities communities. It would be misleading to assume that an observed feature is purely “sycophantic” or “empathic” without due consideration for the context of the personal interaction and the needs of the individual.

7.2 Discussion

7.2.1 Summary of Findings

7.2.2 Future Directions

Chapter 8

Ethics Statement

Attach the ethics document.

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Appendices

Appendix A

An Appendix of Some Kind

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Appendix B

Another Appendix

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